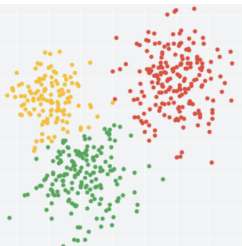


Clustering Evaluation

Procheta Sen



Clustering Evaluation

1. Extrinsic methods (supervised methods)

- ▶ ground truth labels are used
- ▶ assign a score to a clustering given the ground truth labels

2. Intrinsic methods (unsupervised methods)

- ▶ only the partition of the objects into clusters is used
- ▶ examine how well the clusters are separated and how compact the clusters are

Extrinsic methods: evaluate clustering as classification

1. Assign each cluster the label that appears most in that cluster
2. Merge clusters with the same label
3. Measure Precision, Recall, and F-measure for each label type
4. Compute the macro-average, i.e. the average over all label types (classes), of
 - ▶ Precision
 - ▶ Recall
 - ▶ F-score

Extrinsic methods: evaluate clustering as classification

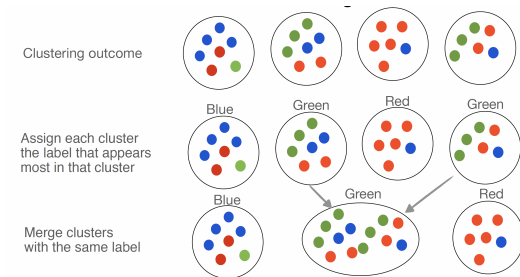


Figure:

Extrinsic methods: B-CUBED Measure

- ▶ Proposed in (Bagga B. Baldwin = B³)
 - ▶ A. Bagga and B. Baldwin. *Entity-based cross document coreference resolution using the vector space model*, In Proc. of 36th COLING-ACL, pages 79–85, 1998.
- ▶ We would like to evaluate clustering without labelling any clusters.

$$\text{precision}(x) = \frac{\text{No. of items in } C(x) \text{ with } A(x)}{\text{No. of items in } C(x)}$$

$$\text{recall}(x) = \frac{\text{No. of items in } C(x) \text{ with } A(x)}{\text{Total no. of items with } A(x)}$$

$A(x)$: label of x

$C(x)$: The ID of the cluster that x belongs to

Extrinsic methods: B-CUBED Measure

- Compute the average over all the items (instances) that appear in all clusters (N)

$$\text{Precision} = \frac{1}{N} \sum_{p \in \text{DataSet}} \text{Precision}(p)$$

$$\text{Recall} = \frac{1}{N} \sum_{p \in \text{DataSet}} \text{Recall}(p)$$

$$F\text{-Score} = \frac{1}{N} \sum_{p \in \text{DataSet}} F(p)$$

Intrinsic methods: Silhouette Coefficient

1. Let $C_1, C_2, \dots, C_k$ be the clusters
2. For object x (assume $x \in C_i$):
 - ▶ $a(x)$: The mean distance between x and all other points in the cluster of x .
 - ▶ $b(x)$: The mean distance between x and all other points in the *next nearest cluster*.

$$a(x) = \frac{1}{|C_i| - 1} \sum_{\substack{y \in C_i \\ y \neq x}} d(x, y)$$

$$b(x) = \min_{\substack{j=1, \dots, k \\ j \neq i}} \frac{1}{|C_j|} \sum_{y \in C_j} d(x, y)$$

$a(x)$ is a measure of how dissimilar x is to its own cluster, a small value means it is well matched.

$b(x)$ is a measure of how badly x is matched to its neighbouring cluster.

Intrinsic methods: Silhouette Coefficient

3. Silhouette Coefficient of x

$$s(x) = \frac{b(x) - a(x)}{\max\{a(x), b(x)\}}, \quad \text{if } |C_i| > 1$$

$$s(x) = 0, \quad \text{if } |C_i| = 1$$

$s(x)$ close to 1 means that the data is appropriately clustered

$s(x)$ close to -1 means that it would be more appropriate if x was clustered in its neighbouring cluster

$s(x)$ close to 0 means that x is on the border of two natural clusters

4. Silhouette Coefficient of the dataset is the average of Silhouette Coefficients of all objects in the dataset

measure of how appropriately the data have been clustered

Intrinsic methods: Silhouette Coefficient

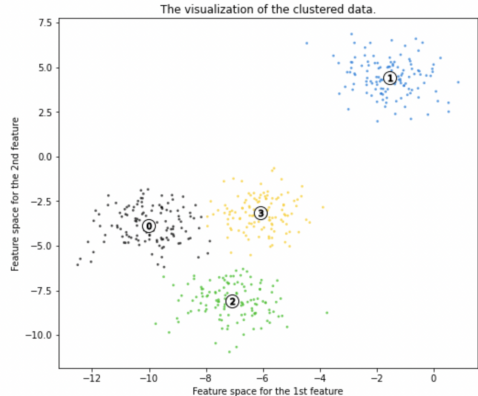
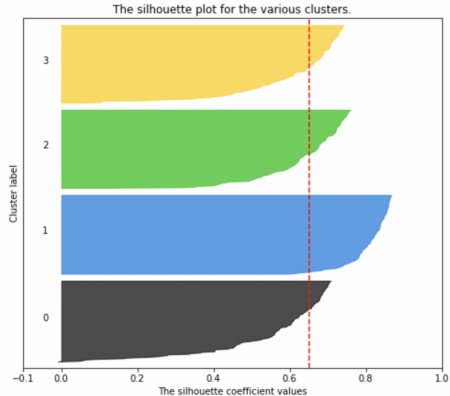


Figure: